

Wookyeong Song

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Research Interests

Object Data Analysis (Interpretable statistical models for complex data in metric spaces), **Metric Geometry** (Manifold learning, latent space), **Applied Statistics** (Casual inference, brain networks, distributional data with optimal transport, and point cloud data), **ML / AI** (Generative modeling).

Education

Ph.D. Candidate in Statistics, University of California, Davis

Davis, CA

Advisor: Professor Hans-Georg Müller

Jun 2026 (Expected)

- Recipient of Julius Blum Award, highest Ph.D. academic award in the department.
- Recipient of Peter G. Hall Award, highest Ph.D. research award in the department.

B.S. in Statistics and Mathematical Sciences (Double Major), Seoul National University

Seoul, South Korea

Advisor: Professor Hee-Seok Oh & Professor Taesung Park

Feb 2021

- Grade: Summa cum laude.

Publications and Preprints

Choi, C.[†], **Song, W.**[†], Müller, H.-G., and Park, B. U. (2026), Additive Fréchet Regression of Random Objects, *Submitted*, [†]co-first author.

Song, W.[†], Zhou, H.[†], Zhou, Y.[†] and Müller, H.-G. (2026), Non-Euclidean Data Analysis With Metric Statistics, *Harvard Data Science Review*, *Accepted*.

Song, W. and Müller, H.-G. (2026), ADOPT: Additive Optimal Transport Regression, *The 29th International Conference on Artificial Intelligence and Statistics (AISTATS)*, *Accepted*.

Song, W. and Müller, H.-G. (2025), Inference for Dispersion and Curvature of Random Objects, *Journal of the American Statistical Association: Theory and Methods*, 1-12, (selected as Student Paper Award Finalist by ASA Nonparametric Statistics Section, JSM 2024).

Song, W., Lim, Y., Cheung, K. and Oh, H.-S. (2023), Multi-feature Clustering of Step Data Using Multivariate Functional Principal Component Analysis, *Statistical Papers*, 65(4), 2109-2134.

Kim, H.[†], **Song, W.**[†] et al. (2023), Development, Validation, and Comparison of a Nomogram Based on Radiologic Findings for Predicting Malignancy in Intraductal Papillary Mucinous Neoplasms of the Pancreas ... , *Journal of Hepato-Biliary-Pancreatic Sciences*, 30(1), 133-143.

Kang, J., Lee, C., **Song, W.** et al. (2020), Risk Prediction for Malignant Intraductal Papillary Mucinous Neoplasm of the Pancreas: Logistic Regression vs Machine Learning, *Scientific Reports*, 10, 20140.

Work in Progress

Song, W., Dubey, P., Petersen, A. and Müller, H.-G. (2026), Inference for Fréchet Regression Effect and its Application to Variable Selection.

Cui, M., **Song, W.**, and Müller, H.-G. (2026), Fréchet Variance Process.

Song, W. and Müller, H.-G. (2026), Intrinsic Geodesic Learning in Metric Ambient Spaces.

Song, W. and Müller, H.-G. (2026), Variance Decomposition of Object-valued Regression in Metric Spaces.

Honors and Awards

2025	Peter G. Hall Award , Department of Statistics, UC Davis	Davis, CA
2024	Best Presentation Award , Special Topic-contributed Session by ASA Nonparametric Statistics Section, JSM 2024	Portland, OR
2024	Student Paper Award Finalist , ASA Nonparametric Statistics Section, JSM 2024	Portland, OR
2023	Travel Award , Princeton Machine Learning Theory Summer School, Princeton University	Princeton, NJ
2022	Julius Blum Award , Department of Statistics, UC Davis	Davis, CA
2020	Excellent Tutoring Award (Mathematical Statistics 1) , Seoul National University	Seoul, South Korea
2014,15,20	Dean's List , College of Natural Science, Seoul National University	Seoul, South Korea
2019	3rd Prize , Poster presentation, Fall Korean Statistical Society Conference	Seoul, South Korea

Conference, Workshop, and Program Participation

JSM 2024, Topic-contributed Paper Session, Portland, OR, USA, 2024

Statistics in the Age of AI, Poster Session, George Washington University, DC, USA, 2024

Princeton Machine Learning Theory Summer School, Princeton University, Princeton, NJ, USA, 2023

Services

Reviewer Journal of the American Statistical Association, Biometrika, AISTATS 2026

Teaching Experience

Associate Instructor

University of California, Davis

Davis, CA

Summer 2025

- STA103: Applied Statistics for Business and Economics (Rating: 4.75/5.0)

Teaching Assistant

University of California, Davis

Davis, CA

- STA010: Statistical Thinking (Fall 2021, Spring 2022)
- STA142A: Statistical Learning I (Winter 2022)
- STA141B: Data & Web Technologies for Data Analysis (Fall 2022)
- STA206: Statistical Methods for Research I (Fall 2022)
- STA100: Applied Statistics for Biological Science (Winter 2023)
- STA013: Elementary Statistics (Spring 2023)
- STA103: Applied Statistics for Business and Economics (Spring 2025)
- STA104: Nonparametric Statistics (Fall 2025)
- STA138: Categorical Data Analysis (Winter 2026)

Mentoring

Muqing Cui (now Ph.D. student at UC Davis)

2023 - Present

- Project: Fréchet Variance Process

Professional Experience

Data Science Ph.D. Intern

Capital One

Plano, TX

Summer 2024

- Developed explainable AI (XAI) models to estimate risk of account-level auto loans, ensuring transparency and optimized performance.

Research Assistant

Multiscale Methods in Statistics Lab, Seoul National University

Seoul, South Korea

Mar 2019 - Feb 2021

- Proposed a robust clustering framework for daily step count data from wearable devices with over 80% of values being zero.

Research Assistant

Bioinformatics and Biostatistics Lab, Seoul National University

Seoul, South Korea

Mar 2019 - Dec 2020

- Built multi-omics hazard model for pancreatic cancer, incorporating gene expressions, somatic mutations, and clinical data points.

Sergeant

Republic of Korea Army

Jeju, South Korea

Feb 2016 - Nov 2017

- Worked as auxiliary police in Korean National Police Agency

Software

frchet Statistical Analysis for Random Objects and Non-Euclidean Data (R package on [GitHub](#)).

References

Hans-Georg Müller

hgmuller@ucdavis.edu

- Distinguished Professor, Department of Statistics, University of California, Davis

Jane-Ling Wang

janelwang@ucdavis.edu

- Distinguished Professor, Department of Statistics, University of California, Davis

Mina Karzand

mkarzand@ucdavis.edu

- Assistant Professor, Department of Statistics, University of California, Davis